Chemistry 3A

Introductory General Chemistry

Experiment 6a

Molecular Modeling

Mitch Halloran

Fresno City College

Introduction

- Molecules which include polyatomic ions can be represented with Lewis structures and ball-and-stick models
- A difference between Lewis structures and the ball-and-stick models is that latter can show molecular geometry
- This experiment will
  - Use ball and stick to represent molecules three-dimensionally
  - Use electron dot Lewis structures to represent molecules two-dimensionally

Background

- Model kits will build molecules in 3-D with balls of different colors for different atoms, and gray connectors
  - Single bonds have short, rigid connectors
  - Double & triple bonds have longer flexible connectors

Table 1: Model Kit Composition

Color	*Element*	*# holes*	*#/kit*
white	hydrogen	1	4
black	carbon	4	3
red	oxygen	2	2
green	chlorine	1	2
orange	bromine	1	2
purple	iodine	1	2
blue	nitrogen *	3	2

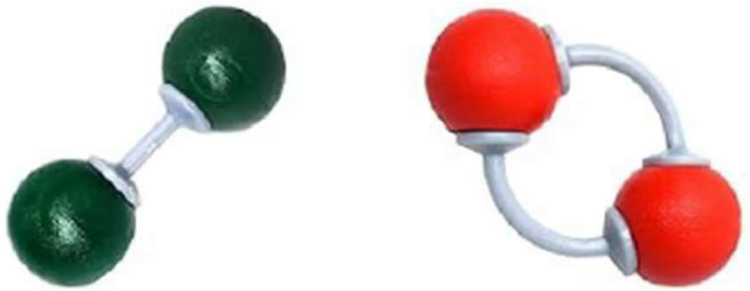
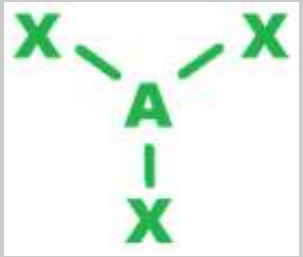
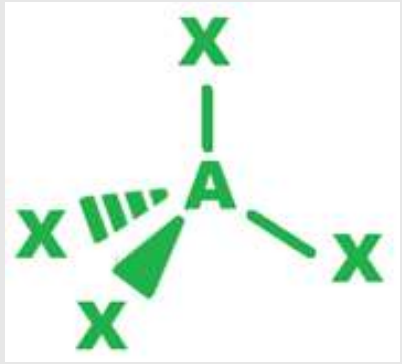
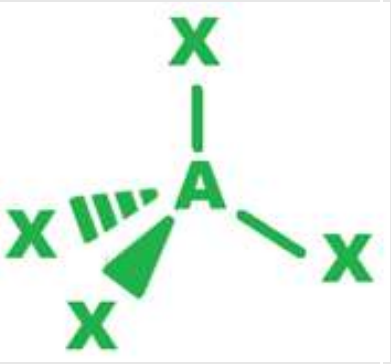
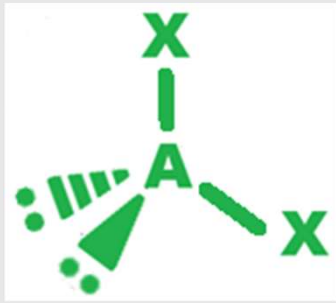


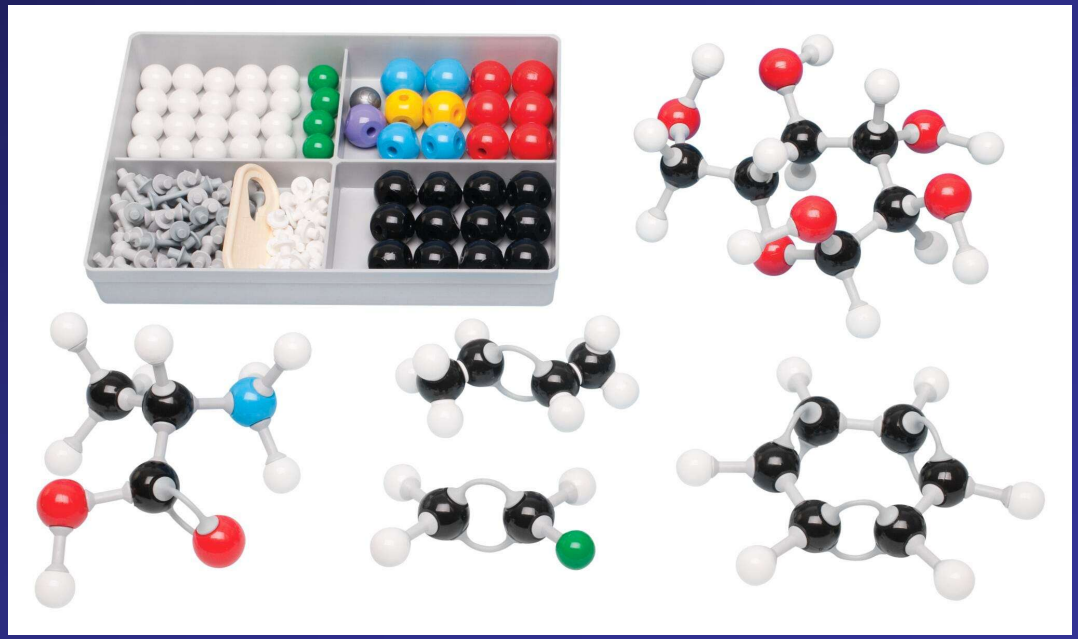
Figure 1. Models showing a single bond (left) and a double bond (right)

Shape Summary

# of Electron Groups	Electron Group Symmetry	Ideal Bond Angle	Molecular Geometry		
			0 lone pairs	1 lone pair	2 lone pairs
2	Linear 	180°	Linear 		
3	Trigonal Planar 	120°	Trigonal Planar 	Bent 	
4	Tetrahedral 	109.5°	Tetrahedral 	Trigonal Pyramidal 	Bent 

Equipment You Will Use





Important Notes

1. This lab report has EIGHT (8) PAGES. So you are to take 8 images and put them in 1 PDF for your report
2. For every **Molecular Model Sketch**, you are to construct the model using the kit pieces. You are then to sketch what you see!!
3. Read the title at the top of the page!
4. "Molecular models with single bonds only"…"with one double bond, with

Procedure

Drawing Lewis Structures First

1. A table with 10 rows representing 10 different molecules will have the Lewis structure drawn first in the lab report
2. Using the molecular model kit, use the correct colored ball and connectors to construct what Lewis structure shows
3. Use colored pencils to sketch the model and color it in in lab report
4. Count electron groups and indicate the geometry of electron groups and molecule. Use appropriate terms to indicate geometry

Procedure

Constructing Molecular Models First

1. A table with 8 rows representing 8 different molecules requires the construction of molecules using modeling kit
2. Complete the lab report with a sketch of molecule from kit components, indicate valence electrons, draw Lewis dot structure
3. Indicate number of electron groups, the geometry of electron groups, and geometry of molecule

Procedure

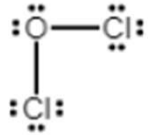
Unknown Molecules

1. Models of five molecules will be presented
2. For each, there are entries to be made in the lab report as follows:
   a. Sketch the model with colored pencils
   b. Determine valence electrons
   c. Draw the Lewis structure
   d. Enter the chemical formula in the 1st column

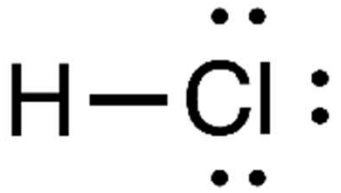
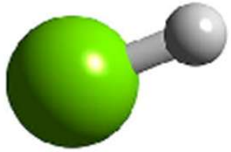
Clean Up

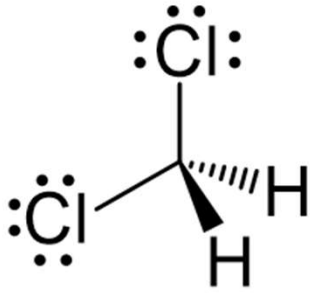
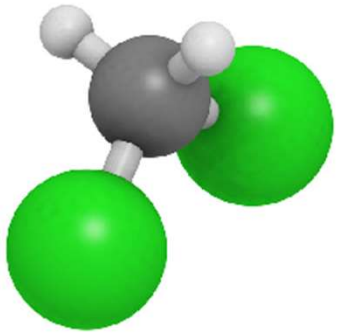
- Return parts to modeling kits
- Organize colored pencils back in storage containers

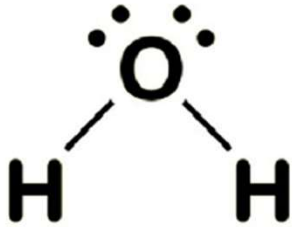
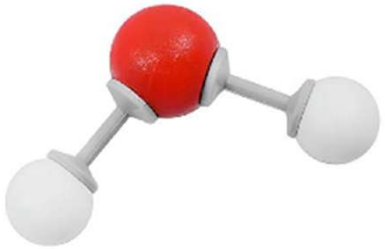
MOLECULAR MODELS WITH SINGLE BONDS ONLY

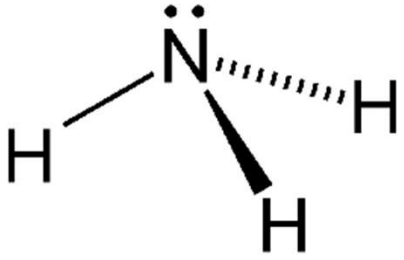
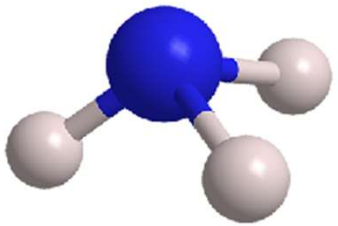
#0	OCl_2		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
20			4 **Electron Group Geometry:** Tetrahedral **Molecular Geometry:** Bent

#1	H_2		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
2	**H—H**		~~X~~ **Electron Group Geometry:** linear **Molecular Geometry:** linear

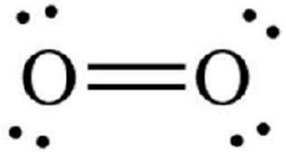
#2	HCl		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
8			~~X~~ **Electron Group Geometry:** linear **Molecular Geometry:** linear

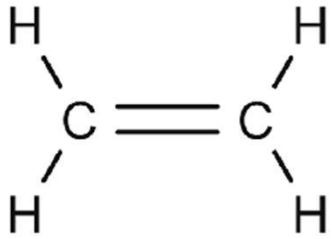
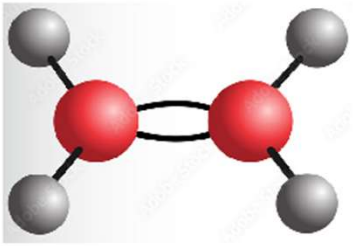
#3	CH_2Cl_2		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
20			**4** **Electron Group Geometry:** tetrahedral **Molecular Geometry:** tetrahedral

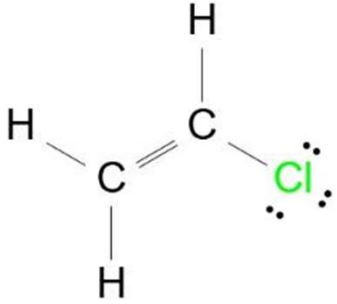
#4	H_2O		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
8			**4** **Electron Group Geometry:** tetrahedral **Molecular Geometry:** bent

#5	NH_3		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
8			**4** **Electron Group Geometry:** tetrahedral **Molecular Geometry:** trigonal pyramidal

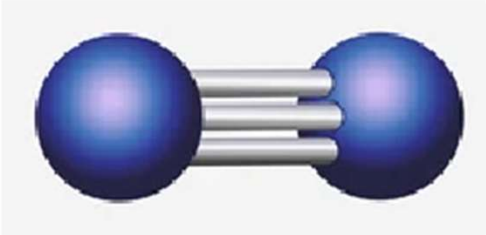
MOLECULAR MODELS ONE DOUBLE BOND

#6	O_2		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
12			~~2~~ **Electron Group Geometry:** linear **Molecular Geometry:** linear

#7	C_2H_4		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups around C:**
12			**3** **Electron Group Geometry around C:** trigonal planar **Molecular Geometry around C:** trigonal planar

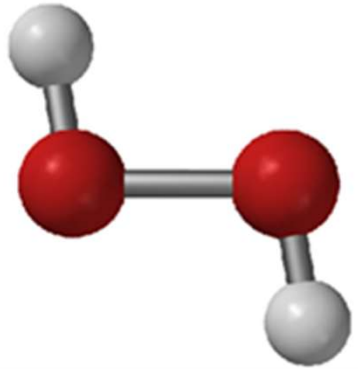
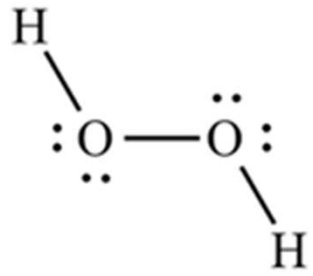
#8	C_2H_3Cl		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups around C:**
18			**3** **Electron Group Geometry around C:** trigonal planar **Molecular Geometry around C:** trigonal planar

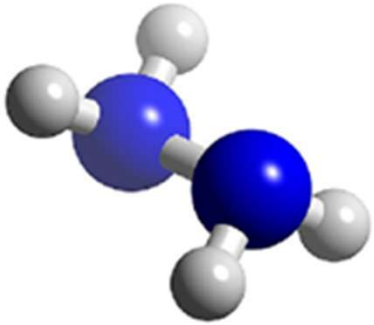
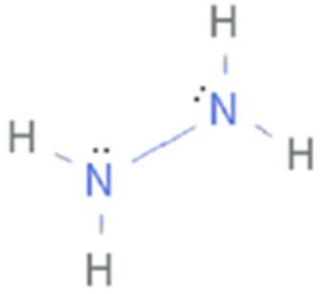
MOLECULAR MODELS ONE TRIPLE BOND

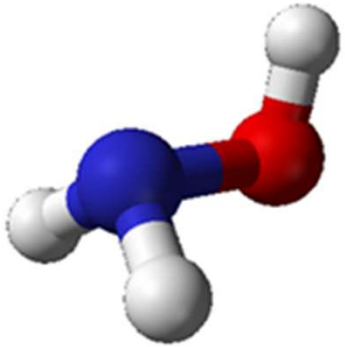
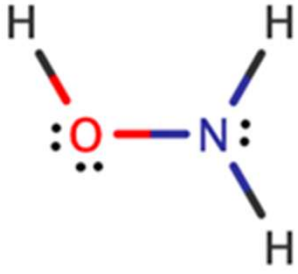
#9	N_2		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups:**
10	:N≡N:		**Electron Group Geometry:** linear **Molecular Geometry:** linear

#10	C_2H_2		
Valence Electrons	**Lewis Dot Structure**	**Molecular Model Sketch**	**Number of Electron Groups around C:**
10	H—C≡C—H		**2** **Electron Group Geometry around C:** **linear** **Molecular Geometry around C:** **linear**

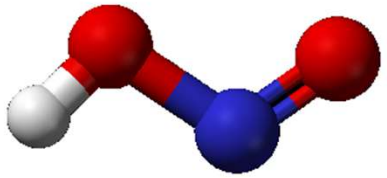
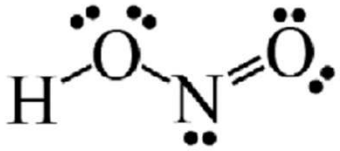
MOLECULAR MODELS WITH SINGLE BONDS ONLY

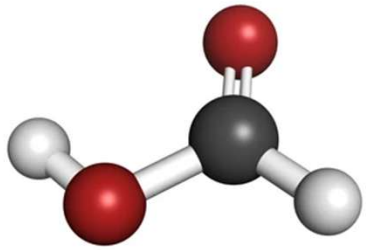
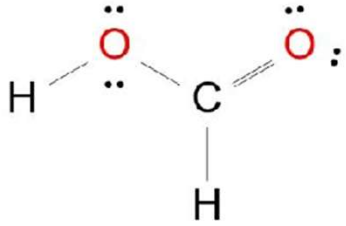
#11	H_2O_2		
Molecular Model Sketch	Valence Electrons	Lewis Dot Structure	Number of Electron Groups around O:
	14		4
			Electron Group Geometry around O:
			tetrahedral
			Molecular Geometry around O:
			bent

#12	N_2H_4		
Molecular Model Sketch	Valence Electrons	Lewis Dot Structure	Number of Electron Groups around N:
	14		4
			Electron Group Geometry around N:
			tetrahedral
			Molecular Geometry around N:
			trigonal pyramidal

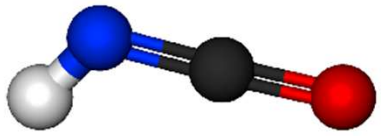
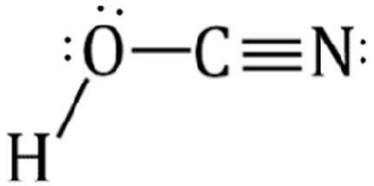
#13	NH_2OH		
Molecular Model Sketch	Valence Electrons	Lewis Dot Structure	Number of Electron Groups around N:
	14		4
			Electron Group Geometry around N:
			tetrahedral
			Molecular Geometry around N:
			trigonal pyramidal

MOLECULAR MODELS WITH ONE DOUBLE BOND

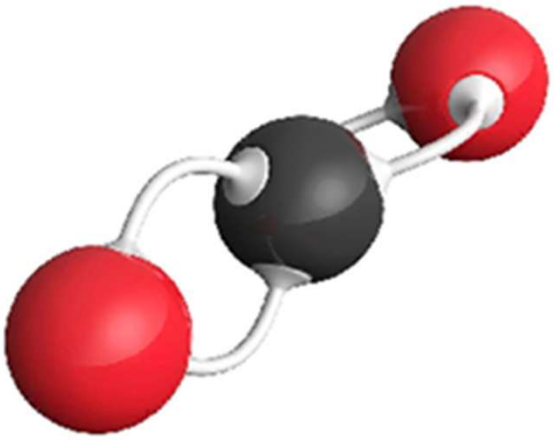
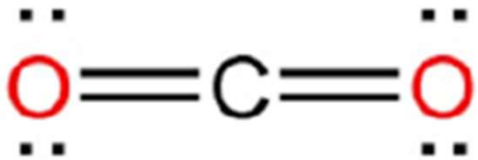
#14	HONO		
Molecular Model Sketch	Valence Electrons	Lewis Dot Structure	Number of Electron Groups around N:
	18		3
			Electron Group Geometry around N:
			trigonal planar
			Molecular Geometry around N:
			bent

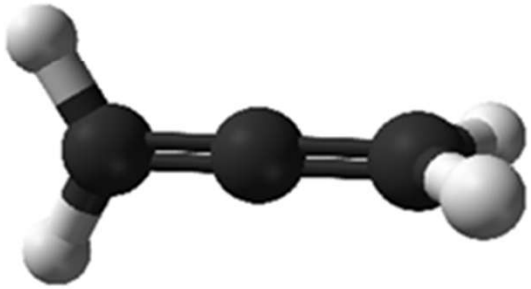
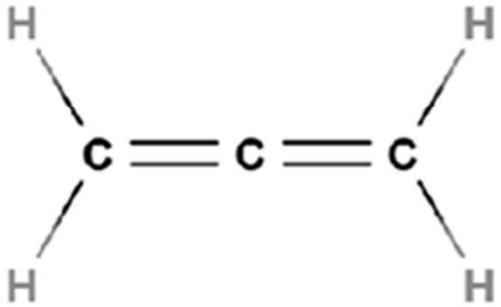
#15	HCOOH		
Molecular Model Sketch	Valence Electrons	Lewis Dot Structure	Number of Electron Groups around C:
	18		3
			Electron Group Geometry around C:
			trigonal planar
			Molecular Geometry around C:
			trigonal planar

MOLECULAR MODELS WITH ONE TRIPLE BOND

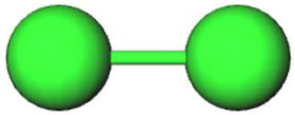
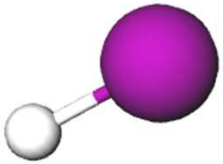
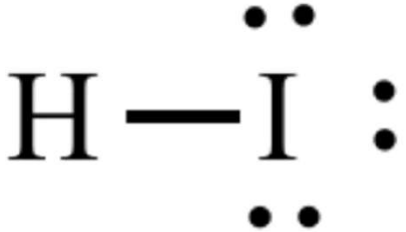
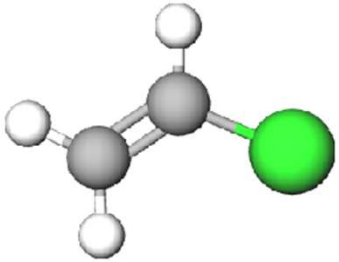
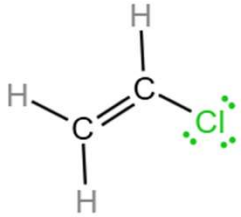
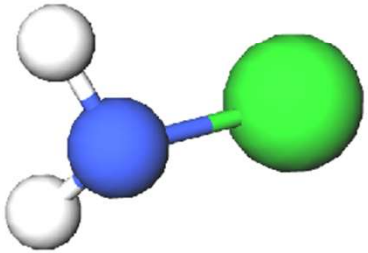
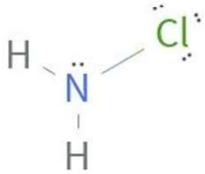
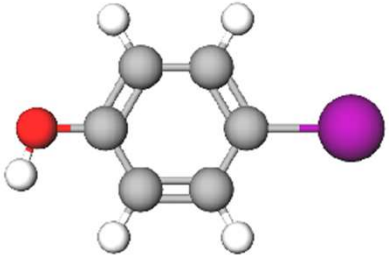
#16	HOCN		
Molecular Model Sketch	Valence Electrons	Lewis Dot Structure	Number of Electron Groups around C:
	16		2
			Electron Group Geometry around C:
			linear
			Molecular Geometry around C:
			linear

MOLECULAR MODELS WITH TWO DOUBLE BONDS

#17	CO_2		
Molecular Model Sketch	**Valence Electrons**	**Lewis Dot Structure**	**Number of Electron Groups around C:**
	**16**		**2** **Electron Group Geometry around C:** linear **Molecular Geometry around C:** linear

#18	C_3H_4		
Molecular Model Sketch	**Valence Electrons**	**Lewis Dot Structure**	**Number of Electron Groups around central C:**
	**16**	 3 isomers but 1 with 2 double bonds	**2** **Electron Group Geometry around central C:** linear **Molecular Geometry around central C:** linear

UNKNOWN MOLECULAR MODELS

Molecule	Molecular Model	Valence Electrons	Lewis Structure
#1 Chemical Formula: Cl_2		14	
#2 Chemical Formula: HI		8	
#3 Chemical Formula: $H_2C=CHCl$		18	
#4 Chemical Formula: NH_2Cl		14	
#5 Chemical Formula: $C_6H_4I(OH)$		42	